

The Nature of Time in Physics, Philosophy and Psychology

Understanding the Concept of Time

-An answer to the question "what is time"

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Abstract:

Time, as one of the most fundamental yet mysterious concepts in human cognition, has long been a core subject of inquiry in physics, philosophy, and psychology. Drawing upon a novel ontological framework—Spatial Geometric Realism and its associated Space Elementary Quantum (SEQ) network model—this article attempts to integrate these three disciplinary perspectives on time. It first argues that any qualified theory of time must answer three fundamental questions: What is the subject of change? What is the driver of change? And what is the distinction between objective change and subjectively observed change? By examining traditional conceptions of time—from Newton, Leibniz, and operationalism to relativity theory—this article identifies their shortcomings with respect to these three criteria. It then proposes a definition of time that satisfies these criteria: global objective time is the discrete sequence of panoramic snapshots of cosmic energy distribution over a fixed topological grid, driven by energy-gradient-directed redistribution; subjective time is the local mapping of this sequence through finite-rate sampling. This framework not only provides an ontological foundation for physics' "time parameter," but also builds a conceptual bridge to philosophy's "experience of time" and psychology's "perception of time," revealing the unity and divergence of time across these three dimensions.

1. Introduction: The Puzzle of Time and the Interdisciplinary

Inquiry

"What is time?" St. Augustine famously remarked: "If no one asks me, I know; if I wish to explain it to one who asks, I do not know." This ancient perplexity captures

the dual nature of time as both foundational and elusive. Time is not only a core variable in physics, but also a cornerstone of metaphysical inquiry in philosophy, and a fundamental dimension of human consciousness and behavior in psychology. Yet it is precisely this interdisciplinary centrality that makes the question of time's essence so profoundly complex.

In physics, time is often treated as a parameter—whether Newton's "absolute time," the space-time dimension intertwined with geometry in relativity, or the external parameter 't' in the Schrödinger equation of quantum mechanics. Its definition frequently relies on the measuring instrument of the "clock," leading to a logical circularity: we define time by clocks, and clocks by time. In philosophy, from Aristotle and Augustine to Kant, Bergson, and Whitehead's process philosophy, time has been explored as motion, inner perception, a priori category, or the becoming of "actual occasions"—but these discussions often remain disconnected from physics' operational definitions. In psychology, time is a dimension of subjective experience, involving perception, memory, anticipation, and the stream of consciousness. Its rhythm is modulated by emotion, attention, and even physiological states, giving rise to a "psychological time" that diverges markedly from physical time.

Can the gap between these three perspectives be bridged? Is there a unified ontological framework that satisfies the rigor of physics, engages with the core concerns of contemporary philosophy of time, and simultaneously provides a physical foundation for the subjective experience of time in psychology? The recently developed "Spatial Geometric Realism" and its extension—the "Energy-Gradient-Driven Computational Universe" model (centered on the Space Elementary Quantum or SEQ network)—offer an affirmative answer. This framework rests on three foundational theses from the source papers: first, time must be freed from the operationalist circularity and defined from ontological first principles; second, any qualified theory of time must explicitly answer three key questions: "What changes?", "How does it change?", and "What distinguishes objective change from subjective observation?" ; and third, the fundamental ontology of the universe is a vacuum with fixed topological adjacency relations, wherein all change can be mapped as the redistribution of energy over a fixed grid, and time is precisely the sequence of panoramic snapshots of this energy distribution, driven by energy gradients.

This article aims to weave together the core insights of these three papers, using the three criteria as a guiding thread, to develop a cross-disciplinary dialogue on the nature of time across physics, philosophy, and psychology. It seeks to articulate a framework for understanding time that achieves ontological clarity, physical computability, and psychological mapping capacity.

2. Three Fundamental Criteria for a Theory of Time

To assess whether any given theory of time qualifies as "adequate," we must first establish the minimum standards it must meet. Drawing on a diagnostic analysis of existing theories, we can distill three essential questions that any serious theory of time must answer:

First, what is the subject of change? If time is the measure of change, the primary task is to specify what is changing. This subject must be ontologically clear and specifiable as a real entity or state, rather than vague notions like "events" or "processes" that lack an underlying substrate. The theory must answer: What is the fundamental reality that persists through change? What property or variable of this reality constitutes "change" itself?

Second, what is the driver of change? Knowing what changes is insufficient; we must also explain how and why the transition from one state to the next occurs. This requires the theory to specify the physical force, gradient, or principle that drives the state transition. This driver must be expressible in terms of measurable physical quantities, avoiding explanations that resort to mysterious or circular "flows."

Third, what distinguishes objective change from subjectively observed change? This is the most frequently neglected yet crucial criterion. The theory must clearly distinguish between objective change—the actual transformation of the universe independent of any observer—and subjective change—the change perceived or measured by an observer with finite sensory or instrumental resolution. Failing to make this distinction conflates epistemology (how we measure) with ontology (what is real), leading to operationalism, where time is reduced to clock readings and objective reality is abandoned.

Measured against these three criteria, Newton's absolute time fails to specify both subject and driver ("equable flow" is circular); Leibniz's relational time, while avoiding circularity, leaves the subject and physical driver of the "order of events" vague; operationalism simply avoids ontology, replacing definition with measurement; relativity, though providing an excellent gravity-driven dynamics (Einstein's field equations), leaves the ontology of its space-time geometry unclear and heavily depends on observers, obscuring objective time; and quantum mechanics, with its wavefunction dynamics, faces the ontology debate of the wavefunction and the issue of time as an external parameter.

3. Time in Physics: From Parameter to Ontology

3.1 The Dilemma of Traditional Physical Conceptions of Time

Physics has been extraordinarily successful in its treatment of time, yet it has largely remained at the operational level. Newton's absolute time provided a uniformly flowing background reference but did not explain its connection to material change.

Relativity revolutionized physics by uniting time with space and gravity, revealing time's relativity, but the ontology of space-time geometry in general relativity—is it a field? A manifold? The geometry of a vacuum?—remains unsettled at the foundational level. Quantum mechanics treats time as a classical external parameter, accompanying the unitary evolution of quantum states, but this is incompatible with the dynamic nature of space-time in general relativity, giving rise to the notorious "problem of time." Operationalism (Bridgman), though pragmatic in experimental physics, is logically circular in defining time as "what clocks measure," and it commits a *pars pro toto* fallacy—stellar nucleosynthesis, galactic evolution, and biological aging are far richer than clock changes, yet they all occur "in time" .

3.2 The Energy-Gradient-Driven SEQ Model of Time

To overcome these difficulties, the SEQ model based on Spatial Geometric Realism offers a novel ontological framework. Its core postulates are:

- **Ontological Substrate:** The fundamental ontology of the universe is a vacuum with fixed topological adjacency relations, constituting an immutable network of Space Elementary Quanta (SEQ). Matter and radiation are excitations of this vacuum.
- **Subject of Change:** The global state of the universe is represented by its energy distribution over this fixed topological grid. Hence, **the subject of change is precisely the total cosmic energy distribution.**
- **Driver of Change:** The driving force behind energy redistribution is the **energy gradient** between adjacent SEQ nodes. Energy flows irreversibly from higher-energy nodes to lower-energy neighbors, in discrete quanta of Planck's constant. This simple rule unifies the gradient-driven logic underlying the diffusion equation, the heat equation, and even Einstein's field equations, providing a microscopic dynamical foundation for the second law of thermodynamics.
- **Definition of Global Objective Time:** Global objective time is rigorously defined as the **discrete sequence of panoramic snapshots of this cosmic energy distribution.** This sequence is absolute because it corresponds to no observer's sampling; it is the logical serial number of the total cosmic state's evolution. It is neither Newton's "uniform flow" nor operationalism's "clock readings," but rather a globalized and physically grounded version of Leibnizian relational time.
- **Definition of Subjective Time:** No physical observer can directly perceive the global snapshot sequence. They can only access local information via finite sampling rates. Hence, **subjective time is the mapping of the local space snapshot sequence by an observer or instrument, based on a finite sampling rate,** with an upper limit set by the Planck frequency. Psychological time-sense further involves cognitive processing, memory integration, and anticipation built upon this mapping process.

This model simultaneously satisfies all three criteria: its subject is unambiguous (cosmic energy distribution); its driver is explicit (energy gradients); and it strictly distinguishes objective from subjective (global sequence vs. local finite-sampling mapping). It avoids circular definitions, accommodates quantum field theory, general relativity, and thermodynamics, and offers a novel ontological origin for quantum probability (the non-uniqueness of local gradient combinations).

4. Time in Philosophy: Reality, Becoming, and Subjectivity

Philosophical inquiry into time has pervaded the history of Western thought. The SEQ model presented here resonates deeply with several key philosophical traditions while offering a physicalized response to their core concerns.

Dialogue with Process Philosophy (Whitehead): Whitehead argued that ultimate reality consists of the becoming of "actual occasions," with time being the rhythm of this becoming. The SEQ model can be seen as a "physicalization" of this insight: each global snapshot corresponds to an "actual occasion"; the generation of the snapshot sequence corresponds to gradient-driven dynamics; and subjective time corresponds to "prehension"—the finite-rate sampling of the past. The key difference lies in the driver: rather than the "lure of eternal objects," the model identifies **energy gradients** as the physical driver of generation.

Advancement of Relational Time (Leibniz): Leibniz maintained that time is nothing but the order of events, possessing no independent reality. The SEQ model agrees that time is not an independent substance, but it concretizes the "order of events" as the **ordered sequence of cosmic energy distribution states**. It extends the ordering relation from local events to the totality of cosmic states, giving relational time a physical content and computational foundation.

Transcendence of Operationalism: The SEQ model explicitly acknowledges operationalism as a valid measurement strategy but firmly rejects it as an ontological definition. It insists on the existence of a "global objective time" independent of measurement, of which subjective/local time is merely an approximation or mapping. This upholds a realist stance: time is the sequence of change, not the record of measurement.

Response to "Time from Information" Approaches: Some theories attempt to reduce time to information accumulation or entropy increase [2]. The SEQ model argues that information is a second-order concept; it must be information about some physical reality—namely, the energy distribution. Grounding time in information risks inverting the causal order. The SEQ model grounds time in **energy distribution and its gradient-driven changes**; information is merely the extractable differences arising from that process. This aligns more consistently with a physicalist monism.

5. Time in Psychology: Perception, Memory, and the Stream of Consciousness

Psychological time is the most subjective dimension of temporal experience. It is not a simple reflection of physical time but is characterized by distortions, expansions, contractions, and directional flow. The SEQ model provides a physical anchor for subjective time while accounting for its psychological properties.

Physical Basis of Subjective Time: According to the SEQ model, subjective time is "the mapping of the local space snapshot sequence by an observer based on a finite sampling rate" [1][3]. Human sensory and neural systems are precisely such "finite-sampling" systems. We cannot directly "read" the global energy distribution; instead, we receive local excitations (light, sound, touch) propagating through the SEQ network and encode them via neural impulses (sampling). Therefore, **psychological time-sense is directly linked to the rate and content of this sampling process.**

Mechanisms of Temporal Distortion:

Time Dilation and Compression: When local space is deformed by gravity (compression/stretching) or high-velocity motion (via the equivalence principle), the resonance frequency of SEQ in that region is modulated, altering the local network update (state transition) rate [3]. Externally, this means the "rate of objective change" in that region is modified. For an observer within that region, their internal "subjective sampling rate" changes accordingly, causing them to perceive a different flow rate of time compared to an external observer. This provides a microscopic mechanism for relativistic time dilation. It also offers a potential explanation for psychological phenomena where "time flies when you're having fun" and "time drags when you're waiting": states of high arousal or intense attention may correspond to higher internal information sampling rates, meaning more "change snapshots" are perceived per unit of external time, subjectively "stretching" duration.

Memory and Anticipation: Psychological time is not determined solely by current sampling; it also depends on the storage of past snapshots (memory) and the prediction of future snapshots (anticipation). In the SEQ model, memory can be understood as traces of past energy distribution patterns (recorded in the SEQ network or in higher-level material structures), while anticipation is a simulation of future paths based on current energy gradients. The stream of consciousness can be conceptualized as a continuous interpretation of the current snapshot sequence, situated between memory and anticipation.

The Essence of Life and Intelligence: Within the SEQ framework, living systems are understood as "spontaneously evolved cumulative entropy maximization path-planning systems within a thermodynamic context" [3]. Through genetic and

experiential memory, organisms store historically efficient entropy-increasing pathway templates. By recognizing the topological features of the current environmental energy distribution via fractal pattern matching, they invoke pathway options that promise the greatest cumulative entropy production over extended time horizons. Intelligence, particularly human intelligence, dramatically amplifies this capacity through externalized memory—writing, tools, computers—enabling far-reaching prospective planning. This implies that the "future" dimension of psychological time has a deep physical meaning: it represents a forward-looking simulation and selection of the irreversible entropy-increasing paths driven by energy gradients. Psychological time is no longer a purely subjective construction, but rather an advanced cognitive function evolved by complex systems to more efficiently dissipate energy and enhance local entropy production within the grand unfolding of cosmic energy evolution.

6. Conclusion: The Nature of Time in Unity and Divergence

By synthesizing the perspectives of physics, philosophy, and psychology, we arrive at a richer, more multi-dimensional picture of the nature of time.

At the ontological level, time is the objective sequence of changes in the universe's energy distribution. The fundamental reality is a vacuum energy network with fixed topology; change is the irreversible redistribution of energy across this network. This definition answers the ontological question "what is time?" while avoiding circular definitions and operationalist evasions.

At the physical level, time is the computable parameter of this sequence, governed by the dynamics of energy-gradient-driven redistribution. Global objective time and local/subjective time are strictly distinguished—the former is the logical total order, the latter the finite-sampling mapping. This framework accommodates and explains the core laws of general relativity, quantum mechanics, and thermodynamics, providing them with a unified ontological foundation.

At the philosophical level, this framework achieves a critical synthesis and transcendence of Newton's absolute time, Leibniz's relational time, and Whitehead's process philosophy. It preserves the objectivity of relations while endowing process with a physical carrier and a clear dynamical driver, charting a "middle path" between realism and relationism.

At the psychological level, subjective time-sense is anchored in physical sampling mechanisms. Its distortions, as well as its functions of memory and anticipation, are interpreted as cognitive strategies evolved by complex systems to maximize cumulative entropy production under energy-gradient driving. Psychological time is no longer a floating subjective illusion, but rather the reflection of, and a participatory tool within, cosmic evolution as it manifests at the level of consciousness.

Ultimately, time reveals a unified ontological root across physics, philosophy, and psychology—the sequence of changes in energy distribution—yet manifests distinct facets at different levels: the precise parameter in physics, the ultimate question in philosophy, and the vivid experience in psychology. The divergence arises from differing perspectives and tools of inquiry. Yet the dialogue and integration across these disciplines are gradually revealing the full truth of time—the most fundamental dimension of our universe.

Concluding Reflection on the Theoretical Positioning of This

Framework

The definition of time proposed in this framework possesses a clear and coherent logical structure. While it does not align perfectly with mainstream theories of time, it offers an alternative perspective at a moment when both philosophical and physical understandings of time remain notably ambiguous. This model does not claim to be a definitive answer, nor does it dismiss the profound insights embedded in existing physical and philosophical traditions. Rather, it provides a distinct vantage point—one grounded in ontological first principles, explicit dynamical drivers, and a rigorous separation of objective reality from subjective observation—that may help to clarify foundational confusions and open new pathways for interdisciplinary dialogue. It is offered not as a replacement, but as a complement and a challenge: an invitation to rethink the very terms of the question "What is time?" from a perspective that prioritizes conceptual clarity, physical computability, and ontological parsimony.

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